

Vikas U Rao

+91 9071108756 | vikasurao2004@gmail.com | linkedin.com/in/vikas-u-rao | github.com/Vikas-u-rao

SUMMARY

Software developer with hands-on experience building backend services, full-stack web applications, automation tools, and AI-enabled systems. Skilled in Python, Java, JavaScript, React, Next.js, FastAPI, PostgreSQL, cloud infrastructure, and CI/CD, with experience designing, testing, and deploying practical software solutions.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, HTML, CSS
Frameworks & Libraries: React, Next.js, FastAPI, Flask, Streamlit, Gradio, spaCy, Presidio, Redis
Core Competencies: REST API Development, Full-Stack Development, LLM Integration, NLP, Data Privacy, Automation, Model Evaluation, CI/CD, Software Testing
Tools: Git, GitHub, VS Code, GitHub Actions, Azure, Cloudflare R2, Google Cloud Platform

EDUCATION

Bangalore Institute of Technology Bengaluru, India
Master of Computer Applications — CGPA: 8.2 Nov. 2025 – Apr 2027

EXPERIENCE

Freelance Full-Stack Developer Remote
Client Projects Jun 2026 – Aug 2026

- Architected a full inquiry and lead management system for a real estate client using **Next.js, Zod, Prisma/Postgres**, and **Cloudflare Turnstile**, with automated email notifications via Resend.
- Delivered and maintained **Next.js/Prisma** real estate websites and **Viewora**, a two-container Azure-deployed eyewear e-commerce platform (Next.js + Express) with **Razorpay** payment integration.
- Implemented a **GitHub Actions CI/CD pipeline** with gated deployments and a **Cloudflare R2** image delivery architecture to improve deployment reliability and reduce backend load.
- Automated product data collection for a Shopify-based client store using Shopify's public JSON API, eliminating repetitive manual data entry.

Pranava MedLegal Solutions (India) Pvt. Ltd. Bengaluru, India
Software Intern Aug 2025 – Dec 2025

- Developed a **Patient Demographics Extractor** achieving around **90% accuracy** by identifying and structuring key patient details from unstructured medical text.
- Contributed to an internal **automation framework** for extracting structured data from PDFs and automating medical worksheet processing.
- Implemented an **evaluation framework and pipeline** to automate testing and validation of extraction stages using an **LLM-as-a-Judge** methodology.

PROJECTS

PII Anonymization Gateway | *Python, FastAPI, Microsoft Presidio, spaCy, Gradio, Redis* | [GitHub](#)

- Engineered a gateway to detect and anonymize sensitive personal information before sending data to external LLM services, with reversible mappings for restoring anonymized responses.
- Integrated **Microsoft Presidio** and **spaCy NLP** to detect 15+ PII categories, including names, emails, payment data, and Indian identifiers such as PAN, Aadhaar, GST, and Passport.
- Developed a **FastAPI** backend with **Redis**-based session management for maintaining PII mappings and stateful de-anonymization across requests.

Autonomous Research Agent | *Python, Streamlit, Ollama/Groq, Gemini API, FastAPI* | [GitHub](#)

- Built a hybrid local-cloud research agent that decomposes complex questions, performs concurrent web searches, and identifies contradictions across retrieved sources.
- Designed a **4-stage pipeline** for decomposition, research, contradiction detection, and report generation with configurable Fast and Deep execution modes.
- Integrated **Gemini API with Google Search Grounding** to generate source-backed reports that can be exported as Markdown or PDF with structured citations.

Resume QR Stamper | *Python, Streamlit, PyPDF, ReportLab, OpenCV, QR Code* | [GitHub](#)

- Built a **Python + Streamlit** application for generating and stamping customizable, scannable QR codes onto existing resume PDFs while preserving the original document layout.
- Implemented **PDF placement visualization** with configurable corner positioning, size, margins, page selection, transparent backgrounds, color customization, and logo embedding.
- Added **URL validation, UTM parameter generation, and batch PDF processing** to automate QR-enabled resume generation for multiple application variants.